

Tyler Hartwell

Software Engineer

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Portfolio: tylerhartwell.com

SUMMARY

Software engineer with a mechanical engineering background focused on building responsive full-stack web applications using React, TypeScript, and Next.js. Experienced developing collaborative real-time tools, interactive visualizations, and UI systems with modern frontend and backend technologies.

EDUCATION

Mechanical Engineering, B.S.
UC San Diego

SKILLS

Languages: **TypeScript, JavaScript, SQL, HTML/CSS**
Frameworks/Libraries: **React, Next.js, Tailwind CSS**
Backend/Databases: **Supabase, Firebase, MongoDB**
Tools/Platforms: **Git, GitHub, Vercel, Netlify, REST APIs**

EXPERIENCE

Collaborative Inventory Logging Platform ([Link](#)) 2025+

Tech: Next.js, TypeScript, React, Tailwind CSS, Supabase, Vercel

- Built **relational data models** for items, lists, memberships, invitations, and shared item management
- Implemented **realtime synchronization** using Supabase Realtime subscriptions
- Designed secure **role-based authorization** using Supabase Row Level Security (RLS) policies and PostgreSQL-backed access controls for private shared data
- Built **image upload and asset management** workflows including validation, preview generation, replacement handling, and responsive rendering through Supabase Storage
- Added optimistic UI state management to **minimize perceived latency**

Interactive Weather Visualization ([Link](#)) 2025+

Tech: Next.js, TypeScript, React, Tailwind CSS, HTML Canvas, External API, Vercel

- Engineered a **custom HTML Canvas chart** for simultaneous display of historical and forecast datasets
- Added touch/mouse timeline scrubbing, snapping, scrolling and filtering for **smooth interactive experience**
- Integrated Open-Meteo API data pipelines to **dynamically retrieve data** based on user-selected locations

Advanced Puzzle Solving & Training ([Link](#)) 2024+

Tech: Next.js, TypeScript, React, Tailwind CSS, MongoDB, Vercel

- Built Sudoku training platform with **database of over 3,000 puzzles** organized by difficulty
- Crafted **automated solving system** for common patterns/strategies to isolate advanced techniques

Collaborative Grocery List Management ([Link](#)) 2023+

Tech: HTML, CSS, JavaScript, Firebase, Netlify

- Implemented Firebase Realtime Database synchronization to support **responsive multi-user collaboration**
- Designed anonymous no-login workflows to **reduce onboarding friction** and simplify shared list access

CAD Technician at SeeScan | San Diego, CA 2018-2023

- Collaborated with cross-functional engineering leadership to **design and construct product prototypes** and validation apparatus
- Created and maintained **complex 3D models**, assemblies, and technical drawings
- Collaborated on the development of SolidWorks **macro scripts saving several minutes per drawing**
- Improved and **wrote procedure documentation**, facilitating comprehensive and reliable process adherence

Engineer at CHC Consulting, LLC | San Diego, CA 2017-2018

- Designed cost-effective fiber optic utility layouts for telecom using **specialized CAD tools** and satellite imagery
- Promoted to quality control lead within 9 months, **overseeing design accuracy** and delivery validation across engineering projects